

Notes: Garicano, Lelarge, and Van Reenen (AER, 2016)
Firm Size Distortions and the Productivity Distribution: Evidence from France

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1 Big picture

- **Question.** How costly are size-contingent labor regulations, and how do they distort the firm size, productivity, employment, and welfare distributions?
- **Setting.** The paper studies France, where many labor regulations become binding when firms employ 50 or more workers.
- **Core empirical fact.** French firm counts show a sharp bunching just below 50 employees and a sharp drop at 50 employees. This break does not appear in the United States comparison.
- **Policy object.** The authors want to recover the tax-equivalent cost of the regulatory threshold, not merely document bunching.
- **Model.** The paper embeds a fixed and variable regulatory cost into a Lucas (1978) span-of-control model in which more talented managers optimally run larger firms.
- **Identification idea.** The bulge below the threshold, the valley just above it, and the downward shift in the post-threshold power-law distribution together identify the fixed and variable components of the regulation.
- **Main estimate.** In the baseline specification, the regulations are equivalent to about a 2.3% variable tax on labor for firms above the threshold.
- **Welfare result.** With fully flexible real wages, welfare losses are about 1.3% of GDP. With French-style partial downward real wage rigidity, the welfare loss rises to about 3.4% of GDP.
- **Distributional result.** The regulation mainly hurts workers and large firms. Small firms can gain because the regulation lowers equilibrium wages while they remain below the threshold and avoid the regulatory burden.
- **Economic takeaway.** Size-contingent regulation can misallocate labor away from high-productivity firms, keep productive firms artificially small, and shift resources toward less productive small firms.

2 Incremental contribution of the paper

- **From generic misallocation to a concrete distortion.** The macro misallocation literature often infers wedges from dispersion in productivity or marginal products. This paper ties a wedge to a specific institutional rule: French labor laws that bind at 50 employees.
- **A structural use of bunching.** The paper does not only show that firms bunch below 50. It uses the shape of the entire distorted size distribution to estimate the underlying cost of the regulation.
- **Variable versus fixed regulatory cost.** Earlier threshold analyses often treated regulation as a fixed cost. This paper shows that the data, especially the downward shift in the power law after the threshold, point to an important variable cost component.
- **General-equilibrium incidence.** The model makes clear that workers can bear part of the regulation through lower equilibrium wages. This is central for both welfare and distributional incidence.

- **Wage rigidity as an amplification mechanism.** The paper shows that the same regulatory wedge is much more costly when real wages cannot adjust fully downward, as in France because of the minimum wage and collective bargaining institutions.
- **Population administrative data.** The authors use FICUS, a population dataset of French firms, rather than a survey or selected sample. This is crucial for seeing the exact firm-size distribution around the threshold.
- **Validation beyond employment.** The paper checks whether the estimated model can predict the output distribution, even though the model is estimated primarily from employment-size information.
- **Robustness and extensions.** The analysis considers alternative employment datasets, a second threshold at 10 employees, industry heterogeneity, firm splitting, capital-labor substitution, Hsieh-Klenow-style comparisons, and a dynamic extension.
- **Practical policy contribution.** The paper provides a transparent way to price regulation as a tax equivalent and to ask who gains, who loses, and how large the aggregate cost is.

3 Data and measurement

3.1 Institutional setting

- The relevant French regulations become substantially more demanding at 50 employees.
- Firms at or above the threshold must set up a works council, create a health and safety committee, report detailed labor-contract information, potentially appoint a union representative, establish profit-sharing arrangements, and face additional obligations in collective dismissals.
- The legal threshold is 50 or more workers; in the model the relevant pre-regulation threshold is written as $N = 49$, because firms can stay at 49 to avoid the rules.
- These regulations plausibly combine administrative, managerial-time, bargaining, reporting, and compliance costs.

Interpretation: The institutional feature is sharp and visible in the data. This sharpness is what makes the threshold useful for quantifying regulatory costs.

3.2 Main firm data: FICUS

- The main dataset is FICUS, administrative fiscal data covering the universe of French firms between 1995 and 2007.
- FICUS includes roughly 2.2 million firms per year.
- The baseline empirical analysis focuses on about 200,000 manufacturing firms, because productivity is easier to measure in manufacturing.
- The main employment measure is the arithmetic average of firm head-counts recorded at the end of each of the four quarters of the fiscal year.

- FICUS also includes balance-sheet information such as value added, capital, investment, wage bill, materials, and industry affiliation.

Interpretation: FICUS is especially well suited for this paper because the regulations apply to the firm, not only to plants, and the dataset observes the full distribution of firms around the threshold.

3.3 Alternative employment data: DADS

- The authors also use DADS, a payroll and social-security reporting dataset with worker-level information.
- DADS provides alternative measures of employment, including head-counts and full-time-equivalent measures.
- These alternative measures are useful because different labor regulations use slightly different legal definitions of employment.
- The same qualitative break around 50 employees appears in these alternative data, though the spike can be less sharp depending on the exact definition.

Interpretation: The alternative datasets show that the threshold pattern is not just a quirk of one administrative definition of employment.

3.4 Productivity measurement

- The model interprets firm size as reflecting managerial ability or productivity.
- The authors estimate TFP using production-function methods, including a Levinsohn-Petrin version of the Olley-Pakes control-function approach.
- TFP is used to check whether firms just below the threshold are unusually productive, as the model predicts.
- The baseline estimates of the regulatory cost primarily use the firm-size distribution; TFP provides an important qualitative validation.

Interpretation: If high-productivity firms are constrained just below 50 workers, productivity should spike around the threshold. The paper finds exactly this pattern.

4 Models and empirical specifications

4.1 Lucas span-of-control model with a regulatory threshold

Managers have ability a . If a manager hires n workers, output is

$$y = af(n), \quad f'(n) > 0, \quad f''(n) < 0.$$

The firm faces a regulatory cost only when it crosses the threshold N :

$$\pi(a) = \max_n \begin{cases} af(n) - wn, & n \leq N, \\ af(n) - w\tau n - F, & n > N. \end{cases}$$

- w is the equilibrium wage.
- τ is the variable labor-cost multiplier above the threshold; $\tau - 1$ is the implicit variable tax.
- F is the fixed cost of regulation above the threshold.
- $N = 49$ in the main empirical application, because firms with 50 or more employees are regulated.

For firms away from the threshold, the first-order condition is

$$af'(n^*) - \bar{\tau}w = 0, \quad \bar{\tau} = \begin{cases} 1, & n \leq N, \\ \tau, & n > N. \end{cases}$$

Interpretation: Better managers run larger firms. The regulation changes the cost of labor for firms above the threshold, so some productive firms choose to remain below the threshold instead of paying the cost.

4.2 Equilibrium cutoffs

The model partitions individuals into four groups:

- workers with ability below $a_{\min}$;
- small unregulated firms below the threshold;
- constrained firms that would like to be larger but choose $n = N$ to avoid the regulation;
- regulated firms that are productive enough to cross the threshold and pay the cost.

The threshold ability a_c satisfies

$$a_c = \frac{w}{f'(N)},$$

while a_r is determined by the indifference condition between staying at N and paying the regulatory cost:

$$a_r f(N) - wN = a_r f(n^*(a_r)) - w\tau n^*(a_r) - F.$$

Interpretation: The key empirical objects are the constrained firms and the regulated firms. Constrained firms create the bulge at 49; regulated firms reveal the variable cost through the shifted distribution above the threshold.

4.3 Theoretical predictions

- A variable regulatory tax lowers equilibrium labor demand and therefore lowers wages.
- Lower wages increase entry by small firms and expand the size of firms below the threshold.

- Some firms bunch at $N = 49$ to avoid the regulation.
- A valley appears above the threshold because some firms that would have located there now stay at 49.
- If the variable cost dominates, regulated firms above the threshold are smaller than they would be without regulation.
- TFP should be unusually high near the threshold because productive firms are compressed into the 49-worker bin.

Interpretation: The model turns the visual bunching pattern into a quantitative mapping from the size distribution to regulatory costs and welfare losses.

4.4 Broken power law and measurement error

The empirical framework assumes a power-law distribution of managerial ability and a decreasing-returns production function:

$$f(n) = n^\theta.$$

This implies that the firm size distribution is also approximately a power law. The regulation breaks the power law around the threshold.

- The slope of the log-log firm-size distribution identifies the power-law parameter β .
- The downward shift in the power law after the threshold identifies the variable regulatory cost $\tau - 1$.
- The bulge below 50 and valley above 50 identify the size of the constrained region and the fixed-cost component.
- The authors allow observed employment to differ from theoretical employment:

$$n = n^*(a)e^\varepsilon, \quad \varepsilon \sim N(0, \sigma^2).$$

- Parameters are estimated by maximum likelihood using the observed firm-size distribution.

Interpretation: Measurement error matters because the data do not show a pure mass point at 49 and zero firms just above 49. The empirical model smooths the theoretical broken power law to match the actual distribution.

4.5 Recovering the tax equivalent

The maximum-likelihood procedure estimates the power-law parameter β , the measurement-error parameter σ , the upper threshold n_r , and a shift parameter T . Given a calibrated or estimated returns-to-scale parameter θ , the variable tax equivalent is recovered from the shift in the post-threshold power law.

- The baseline calibration uses $\theta = 0.8$, based on Basu and Fernald.

- Alternative calibrations use $\theta = 0.85$, $\theta = 0.5$, and $\theta = 0.9$.
- The authors also estimate θ from the TFP-size relationship and from production functions.

Interpretation: The model needs a scale parameter because the same size distortion implies different tax rates depending on how strongly output responds to scale.

4.6 Welfare calculations and wage rigidity

The welfare exercise compares total output in the regulated economy with output in a counterfactual economy without the threshold cost:

$$\Delta Y = Y(N, \tau, F) - Y(\cdot, 1, 0).$$

The paper decomposes the welfare effects into:

- gains for some small firms that enter or expand because wages fall;
- losses from firms constrained at the threshold;
- losses from regulated firms that hire too little labor because their labor cost is higher;
- losses from unemployment when real wages do not adjust fully downward.

Partial wage flexibility is parameterized by α , where $\alpha = 1$ means full real wage adjustment. The baseline French-style partial adjustment is $\alpha = 0.70$.

Interpretation: The same regulatory wedge is not equally costly in all labor markets. When wages can fall fully, workers absorb much of the cost and output losses are smaller. When wages are partially rigid, the wedge creates unemployment and larger GDP losses.

5 Identification and empirical design

5.1 What identifies the regulatory cost

- The raw bunching just below 50 shows that firms respond to the threshold.
- The valley above 50 identifies the set of firms that would have crossed the threshold absent the regulation but instead stay below it.
- The downward shift in the power law above the threshold identifies the variable labor-cost component.
- The width of the distorted region and the size of the bunching help identify the fixed component.
- The recovery of the power law farther above the threshold anchors the counterfactual distribution.

Interpretation: The design is structural rather than reduced-form. It infers the missing counterfactual firm-size distribution from the parts of the distribution that remain approximately power-law outside the distorted region.

5.2 Key assumptions

- The counterfactual firm-size distribution would follow a power law around the threshold in the absence of regulation.
- The regulatory threshold is the key reason for the break at 50 employees.
- Measurement error and adjustment frictions explain why some firms appear just above the threshold despite the theoretical valley.
- Returns to scale can be calibrated or estimated well enough to recover the tax equivalent.
- The Lucas model captures the main equilibrium incidence through wages, entry, firm size, and output.

Interpretation: The estimates are model-dependent. The paper is persuasive because multiple observable patterns - size bunching, power-law shift, TFP spike, and robustness across datasets - line up with the theory.

5.3 Validation and robustness logic

- Alternative datasets and employment definitions show similar breaks around 50.
- Model-predicted output shares are close to actual output shares, even though output shares are not the main estimation target.
- Adding a second threshold at 10 employees does not overturn the main 50-employee results.
- Capital-labor substitution reduces welfare losses somewhat but leaves the main conclusions intact.
- Dynamic adjustment costs can explain a positive mass of firms in the valley without changing the core intuition.

Interpretation: The robustness exercises protect the paper from the concern that the threshold pattern is just measurement error, evasion, or a one-factor labor-only artifact.

6 Tables and figures

6.1 Figures 1 and 2: Firm-size distortions around the 50-worker threshold

Read:

- Figure 1 compares the firm-size distributions of the United States and France. Both distributions look power-law-like, but France has a visible distortion around 50 employees.
- Figure 2 zooms into French manufacturing firms with 31 to 69 workers in 2000.
- The number of firms falls sharply at exactly 50 employees relative to 49 employees.
- This raw discontinuity is the starting empirical fact of the paper.

Economic message: The threshold is economically salient. Firms appear to avoid crossing from 49 to 50 employees because the legal obligations at 50 are costly.

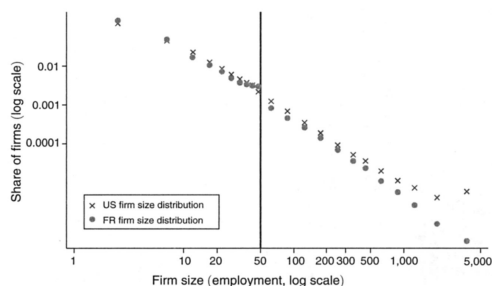


FIGURE 1. FIRM SIZE DISTRIBUTION IN THE UNITED STATES AND FRANCE

Notes: This is the distribution of firms (not plants). Authors' calculations.

Sources: FICUS 2003 for France, and US Census Bureau LBD 2003 for US; population databases of all firms

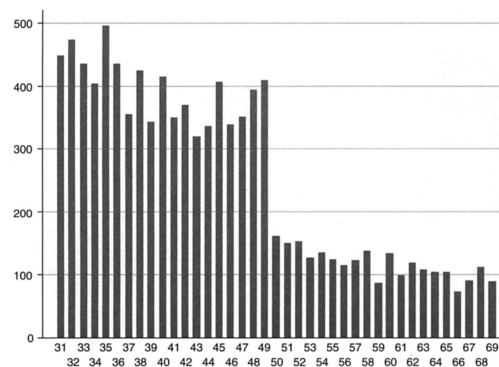


FIGURE 2. NUMBER OF FIRMS BY EMPLOYMENT SIZE IN FRANCE

Notes: This is the population of French firms with between 31 and 69 employees in manufacturing. This plots the number of firms in each exact size category (i.e., raw data, no binning) for the year 2000. There is a clear drop when regulations begin for firms with 50 or more employees.

Source: FICUS

Figure 1: Firm Size Distribution in the United States and France.

Figure 2: Number of Firms by Employment Size in France.

6.2 Figures 3 and 4: Model partition and theoretical predictions

Read:

- Figure 3 shows the model's partition of individuals into workers, small firms, constrained firms, and regulated firms.
- The constrained region is the key region: firms are productive enough to want more than 49 workers but choose to remain at the threshold.
- Figure 4 shows the theoretical firm-size distribution and the theoretical relationship between productivity and firm size.
- The model predicts both a spike in firm mass at 49 and a spike in productivity around the threshold.

Economic message: The model explains why the threshold distorts both size and productivity: productive firms are compressed into the last unregulated size bin.

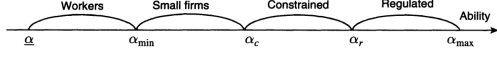


FIGURE 3. EQUILIBRIUM PARTITION OF INDIVIDUALS INTO WORKERS AND FIRM TYPES BY MANAGERIAL ABILITY, α

Notes: This figure shows the definition of different regimes in our model. Individuals with managerial ability below $\alpha_{\min}$ choose to be workers rather than managers. Individuals with ability between $\alpha_{\min}$ and α_c are small firms who (conditional on the equilibrium wage, which is lower under regulation) do not change their optimal size. Between α_c and α_r are individuals who are affected by the regulatory constraint and choose their firm size to be smaller than they otherwise would have been; we call these individuals/firms who are in a constrained regime. Individuals with ability above α_r are choosing to pay the implicit tax rather than keep themselves small.

Figure 3: Equilibrium Partition of Individuals into Workers and Firm Types.

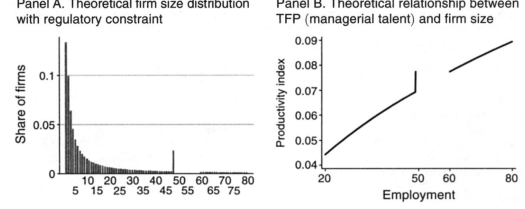


FIGURE 4. THEORETICAL PREDICTIONS FOR FIRM SIZE DISTRIBUTION AND TFP: FIRM SIZE DISTRIBUTION WITH REGULATORY CONSTRAINT

Notes: Panel A figure shows the theoretical firm size distribution; the bar at 49 represents the point at which the size constraint binds. Panel B shows the theoretical relationship between total factor productivity (TFP) and firm size. There is a mass of firms at employment size = 49 where the regulatory constraint binds. Illustrative parameters (selected to amplify the graphical patterns): $\beta_a = 1.6$, $\tau = 1.01$, $n_r = 60$, $\theta = 0.5$, $\beta = 1.3$.

Figure 4: Theoretical Predictions for Firm Size Distribution and TFP.

6.3 Figures 5 and 6: Measurement error and alternative employment definitions

Read:

- Figure 5 shows how measurement error smooths the theoretical broken power law.
- Without measurement error, the model predicts a sharp spike and a zero-density valley. With measurement error, the empirical distribution is smoother.
- Figure 6 shows the 50-worker break using FICUS and DADS employment definitions.
- The break is clearest in FICUS head-counts but remains visible under alternative definitions.

Economic message: The empirical break around 50 is robust. The model's measurement-error component makes the sharp theory usable with real administrative data.

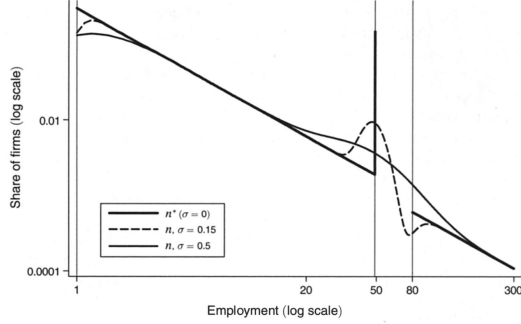


FIGURE 5. THEORETICAL FIRM SIZE DISTRIBUTION WHEN EMPLOYMENT IS MEASURED WITH ERROR

Notes: The thickest solid (blue) line shows the theoretical firm size distribution (broken power law), n^* . The dashed line shows the new firm size distribution when we extend the model, to allow employment size to be measured with error with $\sigma = 0.15$. The solid dark line increases the measurement error to $\sigma = 0.5$. Parameters: $\beta_0 = 1.6$, $\theta = 0.5$, $\beta = 1.3$, but $\tau = 1.02$ and $F/w = 2.9$ such that $n_r = 80$.

Figure 5: Theoretical Firm Size Distribution with Measurement Error.

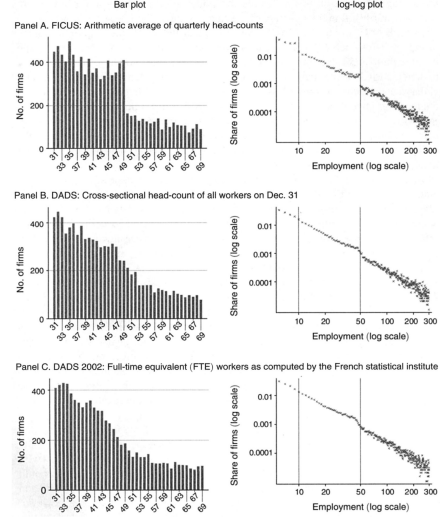


FIGURE 6. THE EFFECT ON THE MEASURED FIRM SIZE DISTRIBUTION USING ALTERNATIVE DATASETS (FICUS AND DADS) AND DEFINITIONS OF EMPLOYMENT

Notes: Data sources are from FICUS (corporate tax collection reported to fiscal administration) and DADS (payroll tax collection reported to social administration); see text and online Appendix C4 for more details on datasets. Panels A and B relate to the year 2000, and panel C relates to the year 2002 (as FTE calculation is unavailable in earlier years). All firms in manufacturing. FICUS employment in head-count at end of quarter averaged over four quarters (panel A). Panel B is head-count on December 31. FTE calculated in panel C by the French Statistical Agency INSEE over the available sample of workers (76 percent of all workers). This excludes jobs lasting less than 30 days or less than 1.5 hours worked per day, or those that were associated with (total) gross earnings lower than three times the monthly minimum wage (using as a benchmark, the seventy-fifth percentile of the series of hours per workers in each two-digit industries and size classes, and are bounded above by 1).

Figure 6: Alternative Datasets and Definitions of Employment.

6.4 Figure 7 and Table 1: Productivity evidence and parameter estimates

Read:

- Figure 7 plots mean TFP by firm size and shows a local productivity bulge around 50 employees.
- This supports the model's claim that some unusually productive firms are held below the regulatory threshold.
- Table 1 reports maximum-likelihood estimates under alternative returns-to-scale assumptions.
- In the baseline with $\theta = 0.8$, the power-law parameter is about 1.8, the upper distorted threshold is about 59 employees, measurement error is 0.121, and the implicit variable tax is 2.3%.
- The fixed-cost estimate is negative in the baseline, implying that the main regulatory burden is variable rather than fixed.

Economic message: The regulatory cost is economically meaningful but not enormous as a tax rate. Its aggregate effect becomes large because it distorts firm size and interacts with wage rigidity.

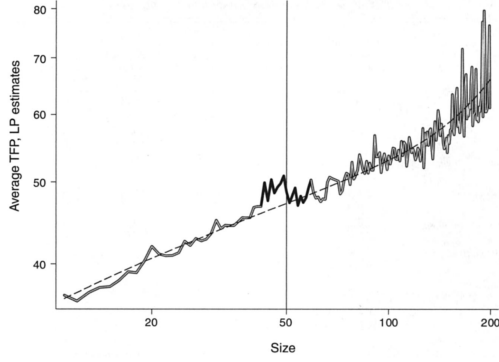


FIGURE 7. TFP DISTRIBUTION AROUND THE REGULATORY THRESHOLD OF 50 EMPLOYEES

Notes: This figure plots the mean level of TFP by firm employment size using the Levinsohn-Petrin (LP) method (see online Appendix B). TFP estimated using an unbalanced panel between 1995 and 2007 of firms having 10 to 1,000 workers. A fourth-order polynomial is displayed using only data from the *undistorted* points (potentially *distorted* points are shown in black). Manufacturing firms only.

Figure 7: TFP Distribution around the Regulatory Threshold.

Method:	θ calibrated: Basu and Fernald (1997) (1)	θ calibrated: Atkeson and Kehoe (2005) (2)	θ calibrated: Hsieh and Klenow (2009) (3)	θ calibrated at 0.9 (4)	TFP/size relationship (5)	Using production function estimates (6)
θ , scale parameter	0.80	0.85	0.50	0.90	0.793 (0.024)	0.860 (0.012)
β , power law	1.800 (0.054)	1.800 (0.054)	1.800 (0.054)	1.813 (0.051)	1.800 (0.058)	1.801 (0.057)
n , upper emp. threshold	59.271 (2.051)	59.265 (2.026)	59.271 (2.052)	52.985 (0.317)	59.271 (2.262)	59.200 (1.715)
σ , measurement error	0.121 (0.033)	0.121 (0.032)	0.121 (0.033)	0.041 (0.003)	0.121 (0.041)	0.120 (0.027)
$\tau - 1$, implicit tax, variable cost	0.023 (0.008)	0.017 (0.006)	0.059 (0.021)	0.007 (0.001)	0.024 (0.008)	0.016 (0.005)
F/w , implicit tax, fixed cost	-0.941 (0.338)	-0.704 (0.252)	-2.375 (0.865)	-0.321 (0.057)	-0.974 (0.345)	-0.655 (0.201)
Mean (median) # of employees	55.8 (24)	55.8 (24)	55.8 (24)	55.8 (24)	55.8 (24)	55.8 (24)
Firms	41,067	41,067	41,067	41,067	41,067	41,067
ln likelihood	-184,128.7	-184,128.7	-184,128.7	-184,196		

Notes: Parameters estimated by ML with standard errors below in parentheses (clustered at the four-digit level). Estimation is on population of French manufacturing firms with 10 to 1,000 employees, in the year 2000. These estimates of the implicit tax are based on different estimates of θ ; the methods are indicated in the different columns. In columns 5 and 6, standard deviations are computed using bootstrap (100 replications). In column 5, the underlying TFP estimates are 0.109 (0.006) for capital and 0.751 (0.014) for labor.

Table 1: Parameter Estimates.

6.5 Figure 8 and Table 2: Cost schedule, model fit, and output validation

Read:

- Figure 8 plots the implied regulatory cost schedule and shows how the model fits the observed firm-size distribution.
- The estimated variable-cost component creates a rising cost above the threshold.
- The model captures the bulge near 49, the valley after 50, and the lower post-threshold power law reasonably well.
- Table 2 compares actual and predicted shares of firms, employment, and output across small, medium, and large firms.
- The model matches firm and employment shares closely and predicts output shares reasonably well, even though output distribution is not the primary target.

Economic message: Table 2 is an important external validation. A model estimated mainly from employment-size patterns also approximates the output distribution.

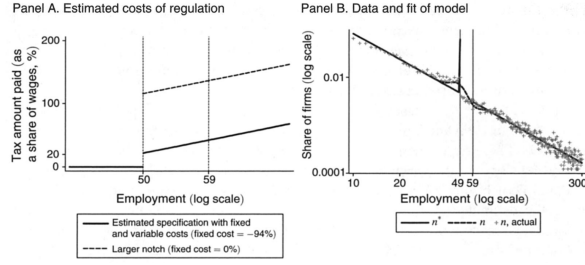


FIGURE 8. IMPLICATIONS OF MODEL ESTIMATES FOR COST OF REGULATION AND FIRM SIZE DISTRIBUTION

Notes: Panel A plots the estimated tax schedule as a function of firm size. For the solid blue line, the estimates correspond to the baseline specification reported in column 1 of Table 1. The dashed red line corresponds to setting the (estimated negative) fixed costs to zero. Panel B shows the difference between the fit of the model (dashed red line, $\hat{n}$) which allows for measurement error with the actual data. Estimates correspond to the baseline specification reported in column 1 of Table 1. We also include the *pure* theoretical predictions (in dark blue solid line, n^*). Actual data (n) are in crosses.

Figure 8: Estimated Cost of Regulation and Model Fit.

TABLE 2—COMPARISON ESTIMATED RESULTS WITH ACTUAL DATA

	Small firms (10 to 48 workers) (1)	Medium firms (49 to 59 workers) (2)	Large firms (60 to 10,000 workers) (3)
Panel A. Firm distribution			
Distribution of firms (actual)	0.761 (0.007)	0.042 (0.002)	0.197 (0.005)
Distribution of firms (predicted)	0.758 (0.002)	0.046 (0.009)	0.196 (0.015)
Panel B. Employment distribution			
Distribution of employment (actual)	0.300 (0.005)	0.040 (0.001)	0.660 (0.005)
Distribution of employment (predicted)	0.276 (0.021)	0.042 (0.008)	0.682 (0.023)
$n = n^*[\alpha] \cdot e^{\epsilon}; \sigma_{\epsilon} = 0.121$			
Panel C. Output distribution			
Distribution of output (actual)	0.249 (0.003)	0.035 (0.001)	0.716 (0.003)
	[0.243; 0.255]	[0.033; 0.037]	[0.710; 0.722]
Distribution of output (predicted)	0.272 (0.021)	0.041 (0.008)	0.686 (0.023)
$y = \alpha \cdot n^{\beta} = \alpha \cdot (n^*[\alpha])^{\beta} \cdot e^{\beta\epsilon}, \sigma_{\epsilon} = 0.121$			
	[0.230; 0.314]	[0.025; 0.057]	[0.640; 0.732]

Notes: Actual distribution computed over our data in 2000 of population of manufacturing firms with 10 to 1,000 employees. Predicted distribution computed using empirical model described in Section II (incorporating measurement error) and our baseline estimates in column 1 of Table 1. Standard deviations for the distribution of actual data are computed from the time variation of empirical distributions over 1995–2007. Standard deviations for predicted values are computed using bootstrap (with 100 replications).

Table 2: Comparison of Estimated Results with Actual Data.

6.6 Figures 9 and 10: Counterfactual distributions and welfare incidence

Read:

- Figure 9 compares the regulated firm-size distribution with the counterfactual distribution without regulation.
- Without regulation, there would be fewer firms stuck below 49 and more firms above the threshold.
- Figure 10 decomposes employment and welfare changes across small, medium, and large firms under full wage flexibility and partial wage rigidity.
- With flexible wages, the aggregate employment loss is small; with partially rigid wages, unemployment rises and employment falls by about 140,000 in the baseline.

Economic message: The counterfactual economy reallocates labor toward larger and more productive firms. Wage rigidity turns what might look like a modest regulatory wedge into a much larger employment and GDP loss.

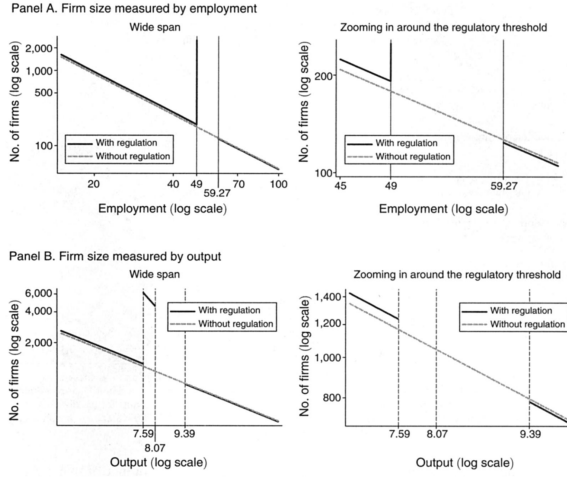


FIGURE 9. FIRM SIZE DISTRIBUTION WITH AND WITHOUT REGULATION

Notes: Figures in both panels compare the firm size distribution in the regulated economy (bold line) from a world without regulation (dashed line) based on the estimated parameters from our model (baseline specification reported in column 1 of Table 1). In the bottom-right panel, the spike has been cut.

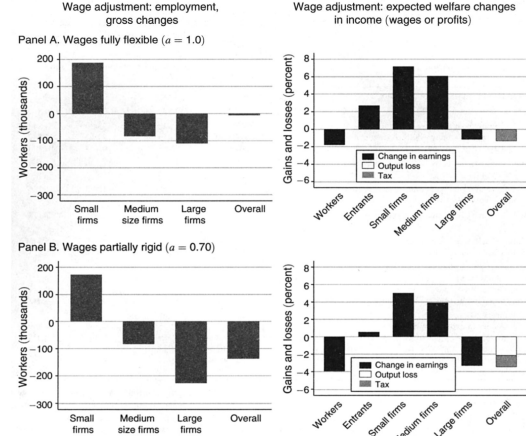


FIGURE 10. EMPLOYMENT AND WELFARE CHANGES BY FIRM SIZE CATEGORY

Notes: These figures illustrate the calculations in Table 3. They are based on the parameter estimates in the baseline model of column 1 in Table 1. Left-hand columns present the change due to the regulation in the number of jobs in each firm size category and right-hand-side figures are welfare changes. Small firms $[\alpha_{\min}, \alpha_L]$ are below the threshold ($N = 49$). Medium firms $[\alpha_L, \alpha_H]$ are larger than 49 but smaller than $n_H (= 59.27)$. Large firms $[\alpha_H, 1]$ are above n_H . Results are taken over 41,067 manufacturing firms (2.3 million workers) with 10 to 1,000 workers in the year 2000 (the range of data used for estimation), employing a total of 2.29 million workers. The right-side figures present percentage changes in earnings for workers and profits of firms in each category (see also Table 3). Workers $[\alpha_L, \alpha_{\min}]$ are individuals who do not choose to set up a firm (employed or unemployed workers) in the economy with regulation. Entrants $[\alpha_{\min}, \alpha_{\min}^W]$ are agents who are workers in the economy without regulation, but entrepreneurs of small firms in the economy with regulation. Small firms $[\alpha_{\min}^W, \alpha_L]$ are entrepreneurs of small firms in both the economy with regulation, and in the economy without regulation. Medium firms $[\alpha_L, \alpha_H]$ are entrepreneurs of (mostly) medium size firms in the economy without regulation, but entrepreneurs of 49 worker firms in the economy with regulation. Large firms $[\alpha_H, \alpha_{\max}]$ are entrepreneurs of large firms in both the economy with regulation, and in the economy without regulation.

Figure 9: Firm Size Distribution With and Without Regulation.

Figure 10: Employment and Welfare Changes by Firm Size Category.

6.7 Table 3 and Figure 11: Welfare, redistribution, and worker valuation

Read:

- Table 3 gives the main welfare and distributional results.
- With perfectly flexible wages, total welfare losses are about 1.326% of GDP.
- With the baseline partial-adjustment parameter $\alpha = 0.70$, the total welfare loss is about 3.450% of GDP.
- Workers lose because wages fall and unemployment appears. Large firms lose because they pay the regulatory cost. Small firms gain because wages fall while they remain below the threshold.
- Figure 11 shows wages around the threshold. The authors do not find a compensating wage drop above 50 that would indicate workers strongly value the regulatory amenities.

Economic message: The regulation has redistributive consequences that are not obvious from the legal text. The main winners are small firms; the main losers are workers and large firms.

TABLE 3—WELFARE AND DISTRIBUTIONAL ANALYSIS

Description of benchmark:	Perfectly flexible real wages (1)	Difference in US/French structural unemployment (2)	Diffusion of min. wage indexation (3)	Standard midterm adjustment assumption (4)
Wage adjustment (degree of flexibility, α)	1.0	0.7	0.67	0.62
θ , scale parameter	0.8	0.8	0.8	0.8
1. Unemployment rate	0	2.622	2.909	3.342
2. Percentage of firms avoiding the regulation, δ	2.920	2.859	2.852	2.842
3. Percentage of firms paying tax (compliers)	10.387	10.168	10.144	10.108
4. Change in labor costs (wage reduction) for small firms (below 49)	-1.792	-1.258	-1.120	-1.109
5. Change in labor costs (wage reduction but tax increase), firms above 49	0.502	1.036	1.095	1.185
6. Excess entry by small firms (percent increase in number of firms)	7.184	7.176	7.175	7.174
7. Increase in size of small firms	8.958	6.292	5.996	5.548
8. Increase in size of large firms	-2.512	-5.178	-5.474	-5.923
9. Annual welfare loss (as a percentage of GDP)				
a. Implicit tax	1.304	1.302	1.302	1.302
b. Output loss	0.022	2.148	2.384	2.741
c. Total (Implicit tax + Output loss)	1.326	3.450	3.686	4.043
10. Winners and losers				
a. Change in expected wage for those who remain in labor force	-1.792	-3.915	-4.152	-4.509
b. Average gain by entering entrepreneurs of small firms	2.667	0.539	0.303	-0.055
c. Average profit gain by small unconstrained firms	7.167	5.034	4.797	4.438
d. Average profit gain by firms constrained at 49	6.06	3.928	3.691	3.333
e. Change in profit for large firms	-1.159	-3.292	-3.529	-3.888

Notes: Based on the baseline estimates in column 1 of Table 1 under the assumption that the maximum firm size is 10,000. Column 1 assumes real wages fully adjust. Columns 2 to 4 assume partial adjustment. Percentage firms avoiding the regulation (row 2) is mass who move to the spike at 49. Percentage firms paying tax (row 3) is mass of agents with productivity greater than α_c relative to agents with productivity greater than α_{min} . Change in labor costs for small firms (row 4) is equilibrium wage effect. Change in labor costs for large firms (row 5) is row 4 plus the estimated implicit tax ($\tau - 1$). Excess entry (row 6) is the difference in the $\ln(\text{mass of agents having productivity greater than } \alpha_{min})$ minus $\ln(\text{mass of agents having productivity greater than } \alpha_{min}^0)$, where α_{min}^0 is the threshold between workers and entrepreneurs in the counterfactual economy without regulation. Increase in size of small firms (row 7) corresponds to $\ln(n^*) - \ln(n_0^*)$ for firms having productivity smaller than α_c ; increase in size of large firms (row 8) corresponds to $n^* - n_0^*$ for firms having productivity greater than α_c . Implicit tax (row 9a) corresponds to the total amount of implicit tax $\int_{\alpha_{min}^0}^{\alpha_c} (\tau - 1) \cdot w^* \cdot n^*(\alpha) + k[\phi(\alpha)] d\alpha$ as a share of total output, Y^* . Output loss (row 9b) corresponds to $\ln(Y^*) - \ln(Y_0^*)$. For winners and losers (row 10), we compute the average (percentage point) changes in expected wages or profits for agents in each of the following bins: 10a labor force $[\alpha_c, \alpha_{max}]$; 10b new entrepreneurs $[\alpha_{min}^0, \alpha_{min}^0]$; 10c. small firms $[\alpha_{min}^0, \alpha_c]$; 10d. constrained firms $[\alpha_c, \alpha_c]$; 10e. large firms $[\alpha_c, \alpha_{max}]$. In the partially rigid wages case in columns 2 to 4, expected wages are computed as $(1 - u_k^*) \cdot w_k^*$ where u denotes the unemployment rate.

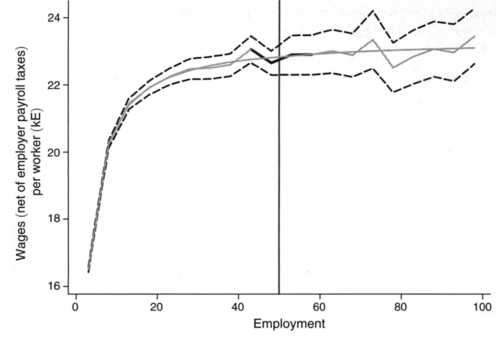


FIGURE 11. WAGES AROUND THE REGULATORY THRESHOLD: ARE WORKERS ACCEPTING LOWER WAGES IN RETURN FOR AMENITY VALUE OF REGULATION?

Notes: Wages is the nominal wage (net of payroll tax) by employer size. Ninety-five percent confidence intervals shown.

Source: FICUS 2000

Table 3: Welfare and Distributional Analysis.

Figure 11: Wages around the Regulatory Threshold.

6.8 Table 4 and Figure 12: Scale-parameter robustness and capital adjustment

Read:

- Table 4 repeats the welfare calculation with a lower returns-to-scale parameter, $\theta = 0.5$.
- The variable tax estimate is larger under this calibration, but the baseline partially rigid welfare loss remains similar in magnitude.
- Figure 12 plots investment per worker around the threshold.
- Investment per worker rises with size and spikes around the threshold, consistent with firms substituting capital for labor when labor regulation becomes costly.

Economic message: The main welfare result is not driven by a single returns-to-scale calibration. Firms also appear to adjust on margins beyond employment, especially capital intensity.

TABLE 4—WELFARE RESULTS: ROBUSTNESS TO CHANGING SCALE PARAMETER θ (from 0.8 to 0.5)

Description of benchmark:	Perfectly flexible real wages (1)	Difference in US/French structural unemployment (2)	Diffusion of min. wage indexation (3)	Standard mid-term adjustment assumption (4)
Wage adjustment (degree of flexibility, α)	1.0	0.68	0.67	0.62
θ , scale parameter	0.5	0.5	0.5	0.5
1. Unemployment rate	0	2.622	2.731	3.137
2. Percentage of firms avoiding the regulation, δ	0.962	0.941	0.941	0.937
3. Percentage of firms paying tax (compliers)	3.420	3.348	3.345	3.334
4. Change in labor costs (wage reduction) for small firms (below 49)	-4.164	-2.833	-2.778	-2.568
5. Change in labor costs (wage reduction but tax increase), firms above 49	1.571	2.902	2.958	3.167
6. Excess entry by small firms (percent increase in number of firms)	6.671	6.667	6.666	6.666
7. Increase in size of small firms	8.328	5.667	5.554	5.136
8. Increase in size of large firms	-3.142	-5.804	-5.916	-6.334
9. Annual welfare loss (as a percentage of GDP)				
a. Implicit tax	1.893	1.891	1.891	1.891
b. Output loss	0.039	1.368	1.424	1.633
c. Total (implicit tax + output loss)	1.933	3.259	3.332	3.524
10. Winners and losers				
a. Change in expected wage for those who remain in labor force	-4.464	-5.490	-5.546	-5.755
b. Average gain by entering entrepreneurs of small firms	-0.017	-1.346	-1.402	-1.611
c. Average profit gain by small unconstrained firms	4.164	2.833	2.778	2.568
d. Average profit gain by firms constrained at 49	3.459	2.128	2.072	1.863
e. Change in profit for large firms	-1.052	-2.383	-2.439	-2.648

Notes: Based on the baseline estimates in column 3 of Table 1, under the assumption that the maximum firm size is 10,000. Column 1 assumes real wages fully adjust. Columns 2 to 4 assume partial adjustment. All definitions are

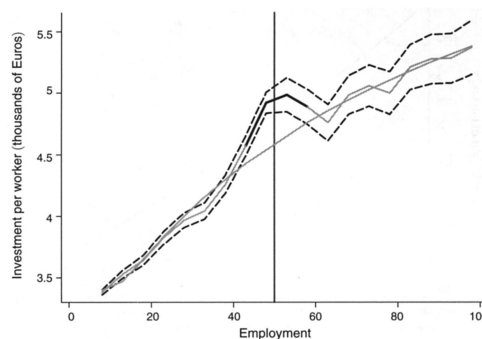


FIGURE 12. INVESTMENT PER WORKER

Notes: Investment per worker, measured in thousands of euros. Ninety-five percent confidence intervals shown. Source: FICUS in year 2000

Table 4: Welfare Robustness to Changing the Scale Parameter.

Figure 12: Investment per Worker.

6.9 Figures 13 and 14: Capital-labor substitution and CES robustness

Read:

- Figure 13 shows how the estimated variable tax changes as the elasticity of substitution between capital and labor changes in a CES extension.
- Higher substitutability lowers the estimated labor tax somewhat, but the estimates remain economically meaningful.
- Figure 14 shows welfare losses under different substitution elasticities.
- Capital-labor substitution reduces welfare losses relative to the baseline but does not eliminate the central conclusion.

Economic message: Allowing firms to substitute into capital weakens but does not overturn the paper's core result. The threshold still distorts employment, firm size, and welfare.

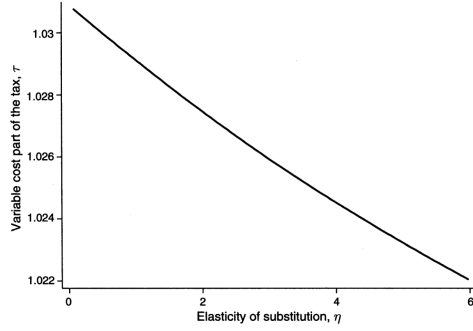


FIGURE 13. ALLOWING FOR CAPITAL/LABOR SUBSTITUTABILITY (CES): HOW REGULATORY TAX ESTIMATES VARY WITH CALIBRATED LEVEL OF ELASTICITY OF SUBSTITUTION

Notes: Parameters estimated by ML. The figure shows how the estimates of the variable cost of the regulation change when we alter the assumptions over the elasticity of substitution (ranging between 0 and 6). See online Appendix D for the relation between estimates of τ in the baseline, single factor model and in this CES specification. Source: FICUS

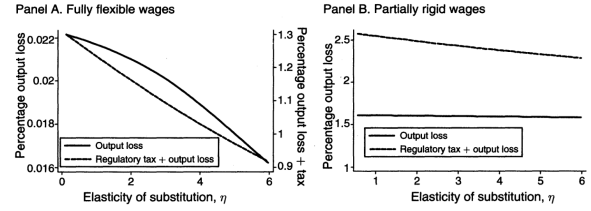


FIGURE 14. ALLOWING FOR CAPITAL/LABOR SUBSTITUTABILITY (CES): HOW WELFARE LOSSES CHANGE WITH ASSUMPTIONS OVER SUBSTITUTABILITY

Notes: Parameters estimated by ML. The figure shows how welfare changes when we alter the assumptions over the elasticity of substitution (ranging between 0 and 6). Panel A looks at the case of fully flexible wages and panel B looks at the case of partially rigid wages, where the adjustment parameter a is calibrated to match the difference in the structural employment rates between France and the United States ($a = 2.622$ percentage points). The price of capital is assumed to be completely flexible in both panels.

Figure 13: CES Substitution and Regulatory Tax Estimates.

Figure 14: CES Substitution and Welfare Losses.

7 Short summary

- The paper studies the welfare and distributional effects of French labor regulations that bind at 50 employees.
- The data show strong bunching below 50 and a sharp drop at 50, especially in the universe of French manufacturing firms.
- The authors build a Lucas-style model in which more talented managers operate larger firms and regulation imposes fixed and variable costs above the threshold.
- The model predicts a bulge at the threshold, a valley above it, a downward shift in the post-threshold power law, and a productivity spike around the threshold.
- These predictions are broadly confirmed in the French data.
- Maximum-likelihood estimation of the firm-size distribution implies a baseline variable labor-tax equivalent of about 2.3%.
- The fixed-cost component is not the dominant part of the estimated burden; the variable-cost component is central.
- With fully flexible wages, the welfare cost is about 1.3% of GDP.
- With French-style partial downward real wage rigidity, the welfare cost rises to about 3.4% of GDP.
- Workers and large firms lose; small firms gain because they enjoy lower wages without paying the regulatory cost.
- The main results are robust to alternative employment data, alternative returns-to-scale assumptions, output-distribution validation, a second threshold, and capital-labor substitution.

8 Conclusion

8.1 Core conclusion

Garicano, Lelarge, and Van Reenen show that size-contingent labor regulation creates meaningful firm-size distortions and aggregate welfare losses. The French 50-employee threshold causes some firms to stay just below the threshold, pushes labor away from more productive firms, and lowers the size of firms that do cross the threshold. In the baseline model, the regulation is equivalent to a 2.3% variable tax on labor above the threshold.

8.2 Economic intuition

The intuition is simple. In a frictionless Lucas economy, high-ability managers hire more workers and run larger firms. A threshold regulation breaks this link. Some productive firms avoid the cost by staying at 49 employees; firms that cross the threshold hire fewer workers than they otherwise would. The result is too much activity in small firms and too little activity in large, productive firms. The economy also adjusts through wages: lower labor demand pushes wages down, which benefits small firms but hurts workers.

8.3 Incremental contribution revisited

The paper advances the literature by turning a visible firm-size notch into a structural estimate of regulatory costs and welfare losses. Its main contribution is not only documenting bunching, but showing how the bulge, valley, and shifted power law identify the implicit tax equivalent of regulation. It also connects micro evidence on a specific legal threshold to macro questions about misallocation, aggregate productivity, and GDP.

8.4 Policy implications

The findings imply that exempting small firms from regulations can create unintended incentives to remain small. This does not mean that all labor regulation is undesirable; the paper does not fully value possible benefits such as worker voice, safety, or workplace representation. But it does show that the costs of threshold-based rules can be substantial, especially when wages are rigid. A smoother regulatory schedule may reduce bunching and misallocation relative to a sharp threshold.

8.5 Limitations

The estimates are structural and depend on the Lucas model, the power-law counterfactual, the measurement-error specification, and the calibrated or estimated returns-to-scale parameter. The paper focuses on France and on the 50-employee threshold, so external validity depends on institutional context. The welfare calculations do not fully quantify all potential benefits or externalities of the regulations. The dynamic model is illustrative rather than fully structurally estimated.

8.6 Takeaway

The enduring takeaway is that firm-size thresholds can have large economic effects even when the implied tax rate looks modest. A sharp regulatory cutoff changes who grows, who stays small, who pays, and who works where. The paper makes the cost of this distortion visible by reading it directly from the shape of the firm-size distribution.